

Ommie: The Design and Development of a Social Robot for Anxiety Reduction

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This article discusses the design, development, and evaluation of *Ommie*, a novel socially assistive robot that supports deep breathing practices for the purposes of anxiety reduction. Research has shown that practicing deep breathing (breathing while extending one's inhales, holds, and exhales) has a strong capacity to calm the autonomic nervous system and reduce anxiety. The robot's primary function is to guide users through a series of deep breaths by way of haptic interactions and audio cues. We utilized a user-centered design approach and present our design methodology in addition to core decisions across robot morphology, tactility, and interactivity. As reported in prior work, the final robot prototype was tested with a two-cohort usability study (n = 43) at a local university wellness center, including participants with anxiety and those with varying levels of experience with deep breathing. Interacting with *Ommie* resulted in a significant reduction in STAI-6 anxiety measures across all participants, who also found the robot intuitive, approachable, and engaging. Participants also reported feelings of focus and companionship when using the robot, often elicited by the haptic interaction. This article describes how our design process and design goals contributed to these results showing *Ommie's* capacity for supporting those with anxiety. Our work also serves as an example of how researchers can design robots for behavioral practices for mental health.

CCS Concepts: • **Human-centered computing** → **Interaction design**; *User studies*; *Haptic devices*; **User centered design**; *Interface design prototyping*; • **Computer systems organization** → **Robotics**;

Additional Key Words and Phrases: social robotics, socially assistive robotics, mental health, anxiety, human-robot-interaction, human-centered design, haptic interactions, robot design, custom robots

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1 Introduction

Anxiety levels are on the rise. In a 2020 CDC study, 25.5% of American adults exhibited anxiety disorder symptoms, three times higher than in the year prior (at 8.1%) [22]. Even prior to the substantial effects of the COVID-19 pandemic, anxiety rates rose by 30% between 2008 and 2018 (from 5.1% to 6.7%) [34], with a 2007 study estimating that 31.2% of adults would experience an

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Fig. 1. A user deep breathing with the *Ommie* robot and receiving haptic guidance as the robot physically expands and contracts. This custom-designed affordance provides the basis for *Ommie*'s interactions.

anxiety disorder in their lifetime [37]. Young adults are being particularly affected by anxiety, with a 2021 National Collegiate Health Assessment showing 27.4% of students having been diagnosed with an anxiety disorder [4]. The individual impact of anxiety goes beyond adverse mood states alone; the disorder has been shown to be negatively associated with work performance, the ability to maintain a healthy social life, and other areas of physical and mental health [95, 96]. Unfortunately, many challenges exist in addressing anxiety rates, including access to treatment (e.g., therapy), cost of treatment, and compliance with behavioral practices [32, 60].

Social robots have a unique capacity to help solve some of these challenges. Rabbitt et al. [73] have outlined the potential of social robots to support mental health by way of distributed and consistent coaching, monitoring, compliance support, and companionship. Past **human–robot interaction (HRI)** work has shown benefits of interacting with an embodied, social agent that may be helpful for mental health, from the ability to motivate new health behaviors [27, 29, 45] to the calming effects of haptic interactions with animal-like robots [75, 101]. Recently, HRI researchers have begun to explore the design of robots specifically for aiding mental health treatment, often through supporting psychological behavioral techniques [5, 11, 40]. However, prior work in this domain is limited in the quantity and scope of studies.

Our work contributes a novel robot design to explore how social robots can support mental health. We focus specifically on generalized anxiety disorder and begin by supporting a well-known and effective behavioral method for anxiety reduction: deep breathing [41]. Deep breathing consists of extending one or multiple of the four breathing phases: inhaling, holding (top of the breath), exhaling, and holding (bottom of the breath). There are multiple configurations that can achieve this extension, from extending inhales and exhales for the same amount of time, to "box breathing," which extends all four breathing phases for the same time. All of these techniques share the same aim: the extensions cause a deeper expansion of the diaphragm from inhaling more air, as well as an overall slower breath. Humans have used deep breathing for centuries, from early Buddhist monks to modern-day meditators, as a path to mindfulness and stress reduction [63]. Recent research has now shown how deep breathing can reduce anxiety through various physiological manners and measures. For instance, the act of deep breathing has been shown to reduce heart rate and cortisol levels [72], two known measures of stress. It has also been shown to calm the autonomic

nervous system [41, 42], an important system in emotion regulation and anxiety. Neurologically, recent research has identified a specific system of neurons that is responsible for both generating breathing rhythms as well as regulating calm and arousal states in an animal model [98]. While more research is to be done, the current state of scientific insight into deep breathing gives credence to its therapeutic benefit for reducing stress and anxiety.

Given the clear beneficial effects of the practice, therapists treating anxiety patients often suggest at-home, daily deep breathing. However, compliance with therapist-given practices is known to be a challenge [32]. Patients can struggle with a number of different parts of these practices, from difficulty remembering to practice, practicing correctly, or sustaining motivation. These obstacles have been addressed with social robots in the past for other skills (e.g., exercising), and a similar approach can thus be taken for anxiety and deep breathing. Social robots have a number of other potential advantages in this domain. By their nature, social robots are able to provide individualized, in-home, in-the-moment support that is not normally possible from a human provider [73]. This type of presence has the capacity to be a complementary extension of mental health support *when* and *where* individuals tend to struggle with behavioral treatments. By providing this support, therapists and care providers can focus less on task performance and compliance and more on deeper, more complex psychotherapy work requiring human connection and conversation. Our aim is not to replace the vital roles that medical professionals and therapists have, rather we seek to develop a tool that can enhance the ease of, and access to, mental health support.

In this article, we outline the design process used to develop the social robot *Ommie* (Figure 1), which supports user mental health through guiding the practice of deep breathing. We describe in detail the iterative steps we took and the design decisions resulting from collaboration with experts, rapid prototyping, and early user testing. These decisions span the robot's initial concept (Section 4), its form factor (Section 5), its breathing mechanism and covering (Section 6), and its digital interactions (Section 7). We additionally describe the full completed robot prototype (Section 8) used in an initial usability study (9) to validate our designs. A condensed description of the robot design and usability study results were originally reported in Matheus et al. [55]. In this article, we provide an artifact of our design process, a detailed explanation of our robot design choices, a deeper description of the end prototype itself, and reflections on how our choices resulted in the findings of the usability study.

2 Related Work

Our work draws from multiple facets of prior HRI and **human–computer interaction (HCI)** research. We particularly draw insights from the field of socially assistive robotics and the application of **socially assistive robots (SARs)**. This sub-field of HRI focuses on how robots can use social interactions to support users with physical, emotional, or mental assistive care needs [28]. For instance, many prior robots have supported individuals with autism spectrum disorder [17], elder care needs [43], and educational challenges [70].

In this section, we describe the prior work in (a) how SARs can support those developing new health skills, (b) how SARs have been used previously in mental health contexts, (c) HCI projects that support deep breathing, and (d) robots that provide therapeutic benefit through haptic interactions.

2.1 SARs for Health Skills Development

Our work seeks to support those with anxiety through not just in-the-moment stress reduction, but through the development of a preventative health skill practiced on a regular basis. SARs have a long history of supporting skills acquisition and behavioral compliance with high efficacy, and coaching and tutoring robots abound [9]. Closest to our work are social robots that support the

development of healthy behaviors and skills related to these behaviors. For instance, robots for supporting exercise and weight loss, in particular, have proven fruitful. Kidd and Breazeal [45] showed that a social robot (in comparison to written logging) can have a measurably strong effect on compliance in calorie tracking and exercise over the course of 6 weeks. Fitter et al. [31] and Fasola and Mataric [27] showed human preference for, and increased engagement with, social robots for motivating exercise with older adults compared to non-robot interventions. Polak and Tzedek [29] also found engagement and motivation effects when using a Pepper robot for rehabilitation exercises with post-stroke patients. Multiple other studies have shown similarly positive impacts of socially assistive robotics for exercise (e.g., [36, 76, 94]).

The many studies showing how SARs are able to support learning new skills encouraged us to design and study a robot for developing a new health skill: deep breathing. Our work builds on this prior work, but it is unique in that relatively few SARs have focused on a mental health skill (see Section 2.2). Our work additionally explores the feasibility and usability of non-verbal haptic interaction, while most existing teaching and coaching SARs use verbal communication through more humanoid forms.

2.2 for Anxiety and Mood Disorders

The design of SARs supporting behavioral treatments specifically for mental health disorders is a small but promising field. Rabbitt et al. [73] detailed the potential applications, benefits, and pitfalls of this domain, including a call-to-action for robots that can be used as a tool for adherence to psychotherapy practices outside of the therapist's office. The authors describe the potential for embodied agents to maintain engagement, provide encouragement, and guide users through self-therapy programs—particularly as a social robot is available on demand, unlike traditional therapy models. Close to our work, Aziz et al. [5] created *CAKNA*, a robot that verbally guides users through psychological techniques, including guided imagery, mindfulness meditation, positive imagery, and deep breathing. *CAKNA* sits on top of a table at approximately 20 inches tall, consisting of a rectangular box with a digital touchscreen tablet in the center and a bear-shaped head on top. The deep breathing feature consists of the robot providing spoken verbal instructions on how to perform deep breathing. In a pilot study, *CAKNA* was shown to reduce anxiety levels more-so than a computer delivering the same content.

Other social robots that support mental health techniques include: *PlantBot* [11], a small, desktop robotic plant with an emoting face that reminds users with depression to perform constructive activities (such as desk cleaning); *Emobie* [3], a fuzzy, portable robot designed for children with anxiety that uses color to project and mirror emotional states; and a *Jibo* robot that delivers positive psychology interventions to college students [40]. While *Plantbot* and *Emobie* have yet to study significant therapeutic benefit, Jeong et al. [40] found that the *Jibo* system provided measurable improvements in psychological well-being after repeated use across seven sessions. They additionally found an improvement in participant readiness-to-change, an important quality for behavior change. The design of this system consists of a tabletop "station" with a Jibo robot and a digital tablet connected by a plastic stand. The positivity psychology interactions included: identifying character strengths, applying strengths to well-being, noting three good things, letters of gratitude, and savoring a small moment.

The work we present in this article contributes a new socially assistive robotics with measurable anxiety state reduction to the growing field of HRI for mental health. While *Ommie* is similar to *CAKNA* and the *Jibo* system as a tabletop robot that guides users through psychological exercises, our robot utilizes haptic instruction and incorporates design features that specifically afford deep breathing.

2.3 HCI for Deep Breathing

While few social robots support deep breathing, several HCI projects have explored the practice through custom-designed haptic devices. For instance, *Breathing Friend* [53] is a small, portable, hand-held device that expands and contracts in the cadence of deep breathing, hoping to help users extend the length of their breaths over time. A study (n = 16) with the device showed that *Breathing Friend* did help participants extend the lengths of their breaths. However, participants reported feeling less relaxed while using the device, with multiple participants commenting on the difficulty of adhering to the device's slower breathing cadence. *PIV* [59] is a wearable system that utilizes vibrotactile stickers placed on the skin on either side of the abdomen. Vibration from the tactors is modulated to guide users through personalized, slow-paced breathing. The system aims to help users regulate their stress inconspicuously throughout the day. Miri et al. [59] showed that using the *PIV* device during the presence of a cognitive stressor led to a decrease in anxiety state measures. However, participants found synchronizing their breath to the vibrations difficult, and the authors found that less effortful interactions with the *PIV* were, in fact, more impactful on reducing anxiety. Taking a different wearable approach, *aSpire* [20] is a belt featuring a series of pneumatic segments that can inflate and deflate in a synchronized or sequential fashion. Yun Choi et al. reported that participants using the *aSpire* on their seat belt while commuting were able to achieve their target breathing rate through biofeedback with the device.

These projects motivated our use of haptic interactions to help instruct deep breathing for anxiety reduction. The design of *Ommie* purposefully addresses the need to make the haptic interaction easy to follow and provide additional breathing cues. While HCI devices are generally more portable than robots, the social relationship that an embodied agent can provide may counter this limitation in its advantages for teaching a new skill or imbuing a sense of companionship that motivates compliance (see Section 2.1). With these potential more social benefits of designing a robot, we chose to pursue an HRI approach rather than build a more general interactive device. Our work seeks to bridge the benefits of SARs with the clear benefits of prior HCI work in deep breathing using haptics.

2.4 Therapeutic Haptics in HRI

Multiple robots have been used to study HRIs that provide calming effects, particularly through haptic interactions [25]. These robots differ from projects in HCI (Section 2.3) and robots like *CAKNA* (Section 2.2) in that they do not seek to provide any instruction in actively modulating one's own breath. Instead, they aim to provide an immediate soothing effect through touch. For instance, *Paro* has been shown by multiple researchers to provide therapeutic benefit through the stroking of a soft robotic seal that moves and purrs [10, 74, 75]. The *Haptic Creature* is another animal-like robot, inspired by research in animal therapy that demonstrates the therapeutic effect of petting live animals. The robot features artificial fur, ear movements, breathing motions, and purring in response to embedded sensors [100]. Sefidgar et al. [80] have shown that holding and stroking the *Haptic Creature* stimulates a relaxation response as measured by both biometric data (e.g., heart rate, galvanized skin response) and self-reporting.

Therapeutic robots with tactilely felt breathing motions are of particular relevance for our work. In addition to the Haptic Creature, two other robots incorporate this feature: (a) *TACO* [64], a huggable robot designed for anxiety and social isolation in children, and (b) *Somnox* [90], a commercially available robot targeting insomnia in adults. Both robots feature abstract geometric forms rather than animal- or human-like ones with internal mechanisms to mimic breathing. While *TACO* has not yet been tested for therapeutic benefit at the time of writing, the *Somnox* team reports that the robot is able to successfully slow down a user's breathing rate while falling asleep [90].

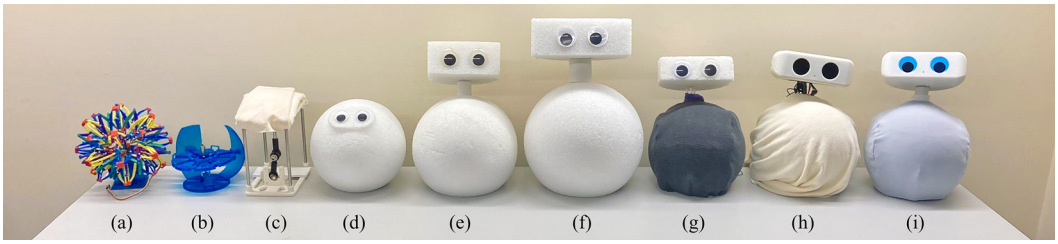


Fig. 2. A selection of mechanical and form factor prototypes developed during the iterative design process to create the *Ommie* robot.

By holding the robot to their chest, users feel its breathing, and their own breathing rate adapts to match. The calming impact of haptic interactions shown through *Paro*, the *Haptic Creature*, and *Somnox* is highly encouraging to our work. We expand research in therapeutic haptics to a semi-anthropomorphized robot that supports practicing a skill (i.e., deep breathing). Past research (e.g., [9, 31, 45]) has shown that these types of anthropomorphized teaching robots can effectively promote the development of health behaviors, as further discussed in the next section.

3 Design Process: Overview

Over the course of 11 months in 2020 and 2021, we utilized a user-centered approach to design and develop the *Ommie* robot. We were particularly inspired by the user-centered processes of **research through design (RtD)** [105] and **design thinking (DT)** [14]. Both approaches share similar goals of building empathy with users, understanding their needs and contexts, and following fast build-test cycles of iteration. The two processes have been popular topics in the HCI community [21, 71, 106, 107] and have been recently discussed from the HRI perspective [51, 52] and applied to a few SARs [8, 33, 56, 85]. Multiple other robots, robot interactions, and robotic concepts have also been created using more general user-centered approaches (e.g., [2, 65, 82, 103]).

Our user-centered design process for *Ommie* took inspiration from RtD, DT, and other user-centered approaches by placing heavy emphasis on exploratory ideation, rapid prototyping, early-stage testing, and multiple rounds of iteration. Figure 2 shows a subset of the prototypes created during our process. Designing the robot began with problem definition, ideation, and context exploration alongside clinical researchers in anxiety (Section 4.1). After selecting the most promising concept, we developed design goals (Section 4.3) and determined the robot form factor through a series of mock-ups and user testing (Section 5). In parallel, we explored various mechanical approaches for the robot's breathing system (Section 6.1) and designed a soft covering for the robot (Section 6.2). Finally, we built and tested digital interactions for the robot's eyes and non-verbal audio cues (Section 7). Our design process resulted in the development of a fully functional prototype (Section 8) for public usability testing (Section 9).[1]

Most new robots are not developed using such an iterative and comprehensive approach [7, 56, 89], and we chose to do so for multiple reasons. For one, we identified early on that no robot on the market today had the affordances we sought for haptic interactions with deep breathing. Thus, a new robot would need to be designed from scratch, requiring a more exploratory approach. Second, we were targeting a vulnerable population that had yet to be robustly studied in the social robotics domain. This necessitated initiating with more early-stage need-finding and expert feedback on this population, as well as iterative testing to ensure our end prototype was appropriate before releasing to a mental health population. Finally, we sought to maximize the potential for positive

[1]All interactions with participants were approved by our local Institutional Review Board.



Fig. 3. Initial ideation sketches of robots that could help reduce anxiety. Left: A robot to support CBT practices. Middle: A robot to support deep breathing support through haptics and audio. Right: Ideas around robot support for mood tracking and daily affirmations.

impact on end-users by the time we reached a fully completed prototype. We accordingly split the system into parts with varying levels of fidelity in testing, based on the design question at hand. This approach is common with DT and other user-centered methods in order to quickly verify smaller choices along the way, leading to larger design impacts.

4 Design Process: Initial Concept

4.1 Problem Definition, Ideation, and Expert Feedback

We began the design process with a simple question: *how might we create a social robot that supports individual mental health?* Initially, we investigated two of the most common mental health conditions, depression and anxiety, through literature reviews to learn their manifestations and the latest treatments. We followed with ideating ways to combine this research with prior knowledge about how social robots can provide haptic calming effects and support skills development. Results of this process included new possible robot interventions, for instance: a robot that helps users practice deep breathing through haptic interactions, a robot that listens to a user's daily affirmations and repeats them back at later times, a robot that routinely checks in with a user on their mental state and provides a listening platform, and a robot that provides interactive **cognitive behavioral therapy (CBT)** cues and rewards.

To select the most promising concept for development, we discussed our ideas with two university researchers with multiple years of combined clinical experience treating anxiety patients. Their research focused on the clinical and emotional manifestations of affect disorders such as anxiety and in predicting anxiety measures neurologically, respectively. Under normal circumstances, we would pair such expert input with qualitative user need-finding. However, this step was adapted, given the ongoing COVID-19 pandemic. During this time, in-person study was necessarily limited, and the mental health population was experiencing a much higher state of vulnerability than normal. In order to respect the ongoing crisis, we utilized the clinical researchers not only as experts but also as proxies for those with anxiety to better understand the context and user needs around our ideas.

Figure 3 shows example sketches from the ideation phase that we brought to these conversations. One of the concepts included a robot that could expand, hold, and contract in the cadence of a series of deep breaths. Users would place their hands on the robot in haptic engagement with the robot's breath, using the cadence as instructions for their own breathing.

The overwhelming response from the clinical researchers was that the robot supporting deep breathing had the highest potential for a number of reasons. First, the clinicians reported that deep breathing is a widely accepted and accessible tool for anxiety reduction in clinical and therapy settings. Thus, it would be easy to integrate with existing care. Second, they described how anxiety patients often have a difficult time conceptualizing how long to inhale, hold, and exhale. This robot could help guide the correct breathing behavior, and the researchers saw the haptic interaction

as simple to follow without the distraction or confusion that verbal instruction can sometimes provide. Third, they were reminded of how tactile interactions have been shown to have calming effects [61, 88, 93], so the haptic interaction was seen as promising in that regard. This feedback solidified our direction into deep breathing support and inspired further conversation on the ideal interaction context for deep breathing with a robot, as discussed in the next section.

4.2 Interaction Context

Given the preference toward a robot for deep breathing support, we explored the potential interaction context and design ecosystem with the same clinical researchers. Focusing primarily on adults and young adults with generalized anxiety, we discussed the following questions with them: (a) what are the current challenges individuals face when attempting to develop a deep breathing practice? (b) what are the suggested guidelines for doing so? and (c) what other types of technologies do individuals typically use?

From the same conversations as in Section 4.1, we learned that while deep breathing exercises are seemingly simple to perform, individuals often have trouble learning or sticking to a deep breathing habit. In particular, user perceptions of their own lengths of inhales and exhales could be incorrect. In the clinics the researchers worked in, users were sometimes guided with verbal counts if they were struggling to keep count on their own. Individuals would generally be instructed to practice the deep breathing exercise daily at home, often in a quiet and private place not associated with stress (e.g., not a work-from-home office, but rather a bedroom or living room while alone). Integrating with an existing routine, such as waking or preparing for bed, could be useful but not required. In addition to this preventative practice, individuals would be encouraged to use deep breathing whenever in an acute moment of stress.

Challenges would commonly arise in remembering to do deep breathing or having the motivation to do so. Often other life needs would get in the way, and individuals might be good at practicing at the start but then proceed to drop off. For some individuals, mobile applications for meditation and deep breathing could be useful, though drop-off is also common with these interactions. Additionally, mobile phones are a known source of stress and anxiety, and well-intentioned users can get distracted by other notifications on their devices. Moving away from the phone and into a technology that could help support grounding through sensory interactions, and support motivation through social-like engagement, was seen as beneficial.

Based on these challenges and guidelines, we imagined the future context of our robotic design as working primarily dyadically in a home setting for a specific individual. This would serve as a "bridge" between any work a user may be doing with a mental health clinician and the preventative habits they are intended to be building at home. However, given that therapy can have barriers to entry, we sought to design a robot that could also be used by an individual in their own pursuit of wellness. Later in development, we also imagined the robot being accessible in community locations such as wellness centers, but early design decisions were made based on an imagined home interaction. This early choice was also influenced by the home-bound nature of many individual's lifestyles during the COVID-19 pandemic.

4.3 Design Goals

We next defined a number of design goals for the tandem breathing concept. These goals were regularly considered during the remainder of the design process and used as anchors for making key decisions across the robot's overall form factor, affordances, and digital interactions.

Social robots inherently have a number of common design elements to consider, such as the robot's trustworthiness, its level of complexity, or approachability. For our design goals, we opted for a small set of the most uniquely suited goals for our use case: reducing anxiety and guiding

a deep breathing practice. The goals were initially developed based on reflection on the problem statement, the discussions with clinical researchers (Sections 4.1 and 4.2), and internal discussion with experts in HRI. Certain goals became clearer during the initial sketching and prototyping process (Sections 5 and 6.1). We discuss the goals and their inspiration below:

(1) *Calming*: The robot should provide a comfortable and calming experience to promote relaxation, wellness, and anxiety reduction. This is particularly important for a population that is more likely be in a high state of stress while interacting with the robot. Accomplishing a calming experience was core to our initial problem statement of supporting individuals with their mental health.
(2) *Engagement*: The robot should inspire use and hold users' attention during interactions, as well as inspire future interaction, in order to support the development of a daily practice. While engagement is a common goal among social robots, we highlighted this need as vital since the best benefits for deep breathing occur when performed as a regular, preventative practice. For the work presented here, we focused first on in-the-moment engagement during the deep breathing session, with the intention that this initial engagement would support future long-term engagement features and interactions.
(3) *Inviting Haptics*: Users should feel comfortable placing their hands on the robot and be able to tangibly feel the robot's breathing. This experience needs to be approachable, pleasant, and trusted to be mechanically robust. This goal was important given that the primary mode of interaction with the robot was to be haptic, and we anticipated most users would not have much prior experience with such an HRI modality. There was a risk that the haptic interaction could be seen as intrusive, awkward, or otherwise strange.
(4) *Breathing Instruction*: The robot should help users perform deep breathing correctly. This means breathing in the correct sequence and lengths of inhales, holds, and exhales for the desired deep breathing cadence. While a seemingly simple exercise, the clinical researchers we connected with described users having trouble applying deep breathing correctly.

In light of these design goals, it was clear that no commercially available robot met all four of them. Heavily mechanized, more humanoid robots, such as *Jibo* [40], *NAO* [35], and *Pepper* [69], may have had high potential for engagement and instruction, but they would have likely failed in providing an inviting haptic or calming experience. In contrast, softer, animal-based robots such as *Paro* [75] or *Keepon* [58] may have provided a calming haptic experience, but they are not designed for instructing or practicing deep breathing skills. Therefore, designing a custom robot became the best way to meet our goal of supporting those with anxiety through practicing deep breathing.

5 Design Process: Form Factor

The start of the robot's physical design began with consideration of its external form factor, as form plays a prominent role in both a user's immediate visual perception of the robot and the physical interactions the robot affords. We focused on three categories of decisions around form factor that would influence our design goals:

(1) *Anthropomorphization*. The question of whether, and how, to anthropomorphize a robot is common to social robot design [24, 30]. Over-anthropomorphizing a robot can risk the *uncanny valley*, where feelings of unease develop around nearly human-like agents [44]. *Perceptual mismatch* is another pitfall to avoid, which occurs when a robot's anthopomorphization is inconsistent across different features, causing negative perceptions by users [44, 68].

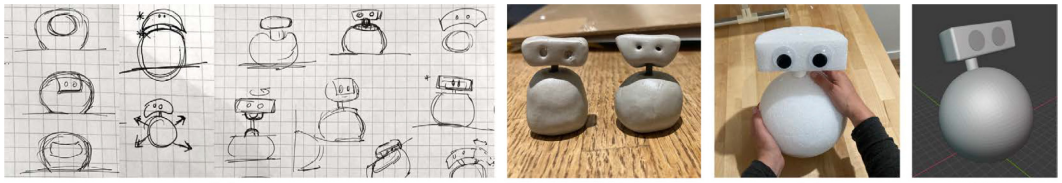


Fig. 4. Examples of the different mediums used in form factor explorations: (a) sketching, (b) small-scale clay models, (c) full-scale foam mock-ups, and (d) digital animation.

(2) *Body and Head Morphology*. Beyond the general level of anthropomorphism, the design of the robot's body and head would have implications on achieving all four of our design goals. Key questions included: how should the robot body be designed to invite the user to place their hands on it? How detailed should the robot's face be, given past studies have shown robot facial features to impact feelings of trust [86]? Where should this face be placed—on a separated head (similar to *CAKNA*), or integrated into the body (similar to *Plantbot*)?
(3) *Robot Scale*. The scale (i.e., physical size) of a robot for deep breathing has many implications. Functionally, there must be enough physical space for a user to place at least one hand on the robot to feel the *haptic interaction*. Perceived agency would also likely be impacted by robot scale. For example, would a smaller robot lack enough perceived agency to *instruct* deep breathing well or provide enough *engagement*? When would a larger robot become intimidating and suppress our *calming* design goal? Finally, the robot's size would impact how much it could be transported around or placed in a lap while performing deep breathing. We saw the potential for such customization of use as beneficial to a mental health population seeking comfort and a quiet space (and differing from past systems such as the *Jibo* system [40] and *CAKNA* [5]).

In order to make these core design decisions, we developed sketches, clay models, and foam mock-ups in increasing fidelity. The results of this design exploration are outlined below and illuminate the thought process behind foundational form factor decisions.

5.1 Form Factor Explorations: Sketches and Clay Models

We first explored anthropomophization and body and head morphology through sketching and clay models, as shown in Figure 4. This approach allowed us to uncover the core questions around these factors early on as well as understand which could be answered early and which would require further prototyping.

5.1.1 Zoomorphic, Anthropomorphic, or Abstract? The form factors of prior social robots that help with health skills or create calming effects vary widely. Robots in the skills acquisition domain typically feature more anthropomorphized forms (e.g., [31, 35, 79]). However, prior robots with breathing motions typically feature abstract geometric or animal-like forms (e.g., [64, 101, 102]). Even purely spherical robots have been shown to elicit lifelike motions and communicate intent [26, 38]. Given these lenses, should our robot be more ball-like and abstract, better facilitating cradling and portability (and thus suspected *haptic* and *calming* effects)? Or, should the robot be more agent-like, with a clearly distinct head and body, likely affording more expressivity for *engagement* and *instruction*?

Through exploratory sketches and in consideration of our design goals, we chose to give the robot a more humanoid than abstract form. Based on the prior work described above, we weighed

the potential for the robot to successfully teach a new skill and maintain engagement as higher with this choice. However, the form would need to balance being an agent that can engage and instruct without the potential discomfort of placing one's hands on an overly human-like robot. This discomfort could be psychological (e.g., feeling awkward at the intimacy of touching a humanoid robot) or physical (i.e., insufficient surface area to place one's hands comfortably). Thus, we sought to design the robot as clearly agent-like but created with as minimal and abstracted forms as possible. This meant the robot need not include human features that did not contribute to the primary deep breathing interaction (e.g., a full face, arms, legs). It also meant that the robot should not be designed with a particular gender in mind.

5.1.2 Smooth Curvatures. Given our design goals, we decided the robot should feature a soft curvature for its body (i.e., bulbous with no hard edges). In the field of neuroaesthetics [19], studies have not only shown that individuals prefer curved objects over sharp ones [84], but that sharp objects activate the neural circuitry of a threat response in the brain [6]. More practically, a bulbous curvature would also provide ample surface area for a user's hands without the need to bend them awkwardly or feel some sort of protrusion. "Palming" the robot generally seemed more approachable to us and less aggressive than "grabbing" the robot. We expected that a spherical-in-nature body would also feel the most natural to the user when feeling the robot breathe. In fact, one of the clinical researchers we spoke to often tells her patients to "imagine a balloon expanding" when instructing deep breathing.

5.1.3 Head Morphologies. Continuing the expanding balloon analogy, should the "balloon" consist of the robot entirely (i.e., an integrated head within a spherical body, as seen in a foam mock-up in Figure 2(d)), or should it be the robot's body with a separate head attached (e.g., Figure 2(e))? Such a choice could have implications on the perceived agency of the robot, with a separated head and neck implying more anthropomorphization than a face on a sphere. Yet, an integrated head would likely make the robot more portable, robust, and easier to handle haptically. At the same time, users might interpret the expanding breathing motion of the robot as if the head was moving in an unnatural or even disturbing manner (i.e., exploding).

At this initial stage, we explored both separated and integrated head approaches, leaving the final decision to future user testing (Section 5.3). For the separated head approach, what head designs to mock up and test became an early design decision. Should the head also be spherical, something rectangular, or perhaps more complex?

Through sketching and clay modeling, we explored a variety of potential shapes and sizes, ultimately settling on a rounded rectangular head to provide a bit of contrast while maintaining soft features. This decision was influenced by the expectation that animated digital eyes would provide the robot with enough facial expression for engagement (therefore eliminating the need for a distinct nose and mouth) [47, 91]. We also deemed the angle and position of the head to be important via sketching. The head should be angled upward in order to have the appearance of looking up at the user for maximum engagement, and the neck should be naturally recessed toward the back of the body, similar to what one would expect from a human body.

5.2 Form Factor Testing: Digital Survey

While sketching and small-scale modeling assisted in categorizing our design decisions and considering the general morphology, we sought to make formal design decisions through user testing with 3D, full-scale mock-ups. However, given the COVID-19 pandemic at the time, indoor in-person studies were not an option early in our design process. Instead, we turned to a digital survey and informal outdoor testing to safely gather early user feedback. The survey featured a number of photos of full-scale foam mock-ups in various home settings (Figure 5) as well as diagrams of



Fig. 5. Select photos from the digital survey showcasing potential ways to use the robot in the home. The left three images showcase the separated head mock-up, the right two feature the integrated head.

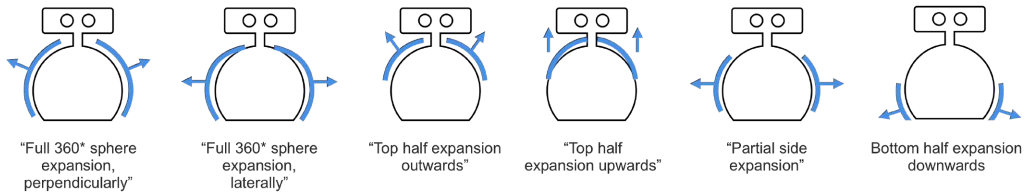


Fig. 6. Vector drawings of possible breathing motion directions for the robot, as used in the digital survey to help guide robot design.

different ways for the robot to physically expand (Figure 6). Respondents were questioned on (a) their preferences of the two head morphologies (integrated and separated), (b) their expectations of how the robot expands, (c) where they would want to place their hands, and (d) where they might want to use the robot in the home.

In total, there were 15 responses from university students. Regarding morphology, 40% preferred a separated head, 33% an integrated head, and 27% found it too difficult to choose. Those preferring the separated head commented that the separation was more intuitive and provided more agency. For instance, one respondent commented, "The separated head makes it easier to see this as an agent rather than a ball." Another stated, "It makes more sense for [the robot] to be breathing from a body vs. a head." Those preferring the integrated head referred to its non-humanoid morphology as more approachable and portable.

For the separated head robot, respondents indicated they would prefer to place their hands primarily on the "shoulders" or the sides of the robot with both hands. The vast majority of respondents imagined the robot expanding outwardly in all directions like a balloon (vs. expanding upward or purely to the sides). They indicated being most likely to use this robot on a desk or tabletop in locations where they feel the most stressed. In contrast, respondents indicated wanting to use the integrated head in their lap on a couch or someplace comfortable. These findings reaffirmed our expectations around the core differences between the two head options. However, we retained the final decision on head morphology for in-person testing with the foam mock-ups (Section 5.3).

5.3 Form Factor Testing: Informal In-Person Testing

While the digital survey provided helpful perspectives, user responses can often differ between one's imagination and reality. Thus, we conducted informal outdoor tests with eight individuals in their 20s and 30s. For the separated-head concept, we tested full-scale foam mock-ups of three different scales and three different robot head shapes. For the integrated head, we tested a smaller and a larger size. Select examples are shown in Figure 7. Users were asked the following questions, or similarly phrased ones, during free-flowing discussion while interacting with the foam mock-ups:

(1) Imagine you were given this robot to practice deep breathing, and the robot itself were to breathe. How do you think it breathes? What part expands?



Fig. 7. Informal user testing with foam mock-ups of varying sizes and form factors. Users preferred the leftmost robot in the leftmost image.

(2) If I were to ask you to place your hands on the robot while it breathes:
   (a) Where do you think you would place your hands, and why?
   (b) How do you feel about it?
   (c) Do you have any reservations about the head getting in the way?
(3) Where might you want to put this robot in your home?
(4) Would you want to bring the robot into your lap at all?
(5) Which of the robot sizes do you prefer, and why?
(6) Which of the head variations do you prefer, and why?

Several findings arose from building and testing the mock-ups live, outlined below.

5.3.1 Body Shape. During the creation of the foam mock-ups, one design element became strikingly clear: a symmetrical spherical body is highly reminiscent of a basketball or soccer ball. These sports balls are often held in the same position we intended for users to place their hands on the robot. This natural hand positioning of the spherical body finalized the question around the exact body form factor from the initial sketching process (i.e., should the sphere have a flat back, like a human does?).

5.3.2 Head and Body Integration. When asked for their preference, all eight users indicated they preferred the separated head. They described a desire for a more "humanoid" morphology and that the purely spherical robot would "inflate and deflate" more dramatically and in unnatural ways. These responses bolstered the results from the digital survey, which had indicated a preference from users for the separated head due to its more intuitive and natural appearance. In this way, we finalized the decision to move forward with a separated head and body.

5.3.3 Hand Placement and Robot Bodily Autonomy. We had initial concerns about large-handed users fitting their hands around the smallest robot, or possibly feeling like they were choking it. However, testing with individuals with large hands did not lead to reported feelings of choking the robot. Interestingly, multiple users noted the intimacy of the interaction instead. One user stated in regard to placing their hands on the mock-up: "It does feel somehow a bit familiar…almost like a hug." Another user commented, "These are styrofoam models with googly eyes, but there was a brief consent moment [before placing my hands on the robot]." This last comment was striking as it raised the concept of robot bodily autonomy. We realized that users might feel like they are intruding upon the robot's personal space or crossing a line when placing their hands on its body. Designing the robot's interactions in a way that gives implicit permission would be important. Relatedly, nearly all users instinctively placed their hands on the robot's shoulders, and many

used the term "shoulders" unprompted. We suspected that the "shoulder" area felt like a safe, close-contact place to touch an anthropomorphized agent.

5.3.4 Imagined Breathing. Distinct from the digital survey, users were split 50/50 on how they imagined the robot breathing between full spherical expansion and the "belly" of the robot expanding straight out. We had not previously imagined the latter option.

6 Design Process: Breathing Mechanics and Covering

In parallel to form factor explorations, we explored different ways to implement the robot's breathing effect. This experience needed to be reliable, smooth, safe, and easily implementable, given its place as the primary interaction modality. Additionally, prior research has shown that users tend to perceive a robot's emotional state based on viewing its breathing patterns [102].

The breathing system consists of both the internal mechanics and the external wrapping, and designing the system centered around three core questions:

(1) What direction should the robot breathe in?
(2) What type of mechanism will provide breathing motions in the most robust and seamless way (i.e., perceived to be natural)?
(3) How can we design the exterior quality of the robot's moving body for a safe, inviting, and haptically pleasant experience?

Below, we outline a number of mechanical system prototypes and experiments with robot clothing that helped us answer these design questions.

6.1 Breathing Mechanism

6.1.1 Mechanism Prototyping. Our goal for the robot's breathing mechanics was for it to not only be significant enough for a user to physically feel (contributing to our *instruction* design goal) but also to appear and feel smooth, natural, and non-threatening (contributing to our *calming* goal). Creating these sorts of fluid, organic motions with mechanical systems is a challenge, but multiple social robots have done so successfully in the past. For example, the *Keepon* robot uses a deformable silicone rubber shell over a system of four motors to rhythmically dance [58], while the *Blossom* robot uses a platform system connected by tensile cables and rubber bands driven by small motors, enclosed with a loose fabric covering [87].

A core design decision for the breathing system was in what direction the robot's breathing should move. We explored two low-fidelity prototypes (Figure 2(a) and (b)) to test outward spherical expansion in all directions (like a balloon expanding), given that so many users had imagined the robot breathing in this manner. However, these systems were mechanically complex and posed a high risk of breaking. This complexity could also translate to a user feeling the internal mechanisms more and thus breaking the *calming* and *haptic* experiences. Accordingly, the most straightforward and robust mechanism was to instead move a curved plate up and down via a constrained plunger on a motor (see Figures 2(c) and 8(c)). The dilemma here was that users had previously expressed concerns about this type of motion seeming unnatural or being misinterpreted (e.g., stretching vs. breathing). To test if the most simple mechanism could still read as natural, we implemented a digital animation comparing the two types of motion. When observing this digital mock-up, we were surprised to see that we perceived the motion, when covered with an expandable material, as expanding partially outward in addition to upward. This observation gave us the confidence to move forward with the simplicity and reliability of the plunger system.

6.1.2 Motor Selection. One concern we had with the breathing system was noise. Would the motor and mechanism's inherent sound inhibit our *calming* design goal by distracting users from

their deep breathing, or presenting the robot as more mechanical than approachable? A side benefit of choosing a mechanically simple mechanism (Section 6.1.1) was reducing the number of moving parts that could produce noise. However, the choice of motor would still have an impact.

During prototyping, we had initially experimented with brushless DC motors, which are known to be near silent and are used in many consumer electronic devices for this purpose. However, these motors are more costly and challenging to encode than the servo motors more commonly utilized in robotics research. To ensure easy controllability, particularly at the slow speeds required for deep breathing motions, we opted to use Dynamixel servo motors despite their typical servo motor sound. The robot's plastic shells and our custom covering (Section 6.2) helped to dissipate this noise, and the motors moved slowly enough to avoid a high-pitched sound. Internal testing of the full prototype, as well as commentary from our usability study, showed that the sounds from the motor mechanism were ultimately acceptable and non-intrusive.

6.2 Robot Covering

In accordance with our *calming* goal, we decided early on that the robot should be soft in nature, especially where users place their hands. Prior research on robot textures shows that users have a strong preference for soft textures on robots [57]. Soft robot coverings exist in many approaches among social robots. Robots such as *DragonBot* [83] and the *Haptic Creature* [101] utilize custom-sewn fuzzy exteriors to mimic animal fur. *HERA*, a haptic robot for children with autism, wears a zippable koala outfit over a *NAO* robot [15], and *Blossom* features loose-fitting handmade woven coverings to maximize robot expressivity [87]. Other robots feature soft, but non-textile, materials such as *Keepon's* silicone rubber exterior [58]. Few custom-designed robots feature custom-made coverings meant to be perceived as clothing. Most robots with clothing are commercial humanoid robots with an off-the-shelf T-shirt or sweatshirt placed on top (e.g., [12, 27]).

For the *Ommie* robot, we chose to design a custom-tailored piece of clothing over a stretchy covering (i.e., skin) rather than covering the robot with fur or human clothing. We made this design decision based on multiple factors. First, given the intimate nature of the haptic experience as described by users in informal mock-up testing, we deemed it important for the robot to not appear naked (meaning the outer covering should not be perceived as skin or fur). Second, given the robot's form factor as only minimally humanoid, using human clothing risked perceptual mismatch [44, 68] by falsely implying the robot's interactions were more human-like than in actuality. Third, we wanted the robot's clothing to be optimized for our *instruction* goal, meaning that users should be able to perceive the breathing motion clearly, both visually and tactilely. This meant the clothing needed to be relatively form-fitting in order to showcase the robot's breathing, which would require a custom design as well. We elaborate below on certain design choices for the robot's covering, which was created in collaboration with a textiles designer.

6.2.1 Skin Creation. We began by developing a soft silicone covering for the robot, overlaying the moving parts of the robot's body while breathing. This skin accomplished two purposes: (a) providing a softness to the outer feel of the robot instead of the feeling of hard plastic, and (b) bridging the gaps in-between the plates of the robot's body (Figure 8(b)) in order to minimize a disjointed appearance and feel while breathing. Early testing with fabric alone showed this gap to be quite visible (Figure 8(c)), which could be perceived as unnatural or disconcerting to users. Utilizing a thick silicone skin under the sweater fabric made this gap less pronounced.

The skin was created by casting with Smooth-On ECOFLEX 00-35 fast-curing silicone, based on prototyping with multiple materials. The molds were custom designed in CAD software, and the skin design features interlocking features with a set of clamps to hold the silicone in place while the robot is breathing (more information in Section 6.2.4). Small tests, similar to those seen

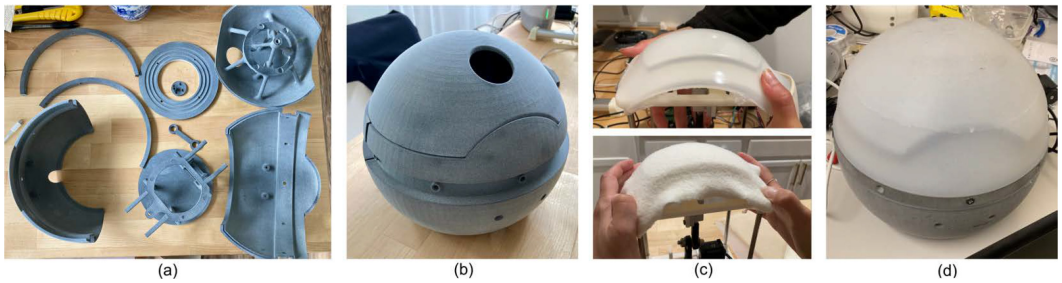


Fig. 8. Prototyping the robot's breathing mechanism and skin covering. (a) 3D-printed body shell components; (b) assembled shell, showing the seam between the moving top plate and the remainder of the body; (c) testing the stretch and visual quality of different covering materials; (d) the final silicone covering clamped to the robot's body, shown with the top plate extended.

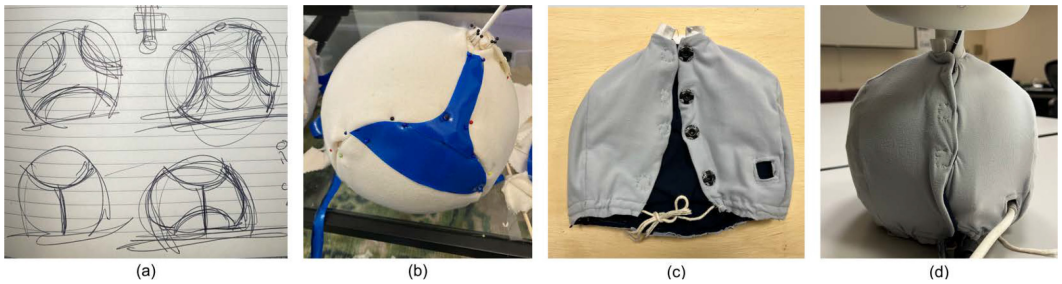


Fig. 9. Prototyping the robot's clothing. (a) Sketches of how to form the fabric into a sphere; (b) experimentation with grip materials; (c) the final robot "sweater" made of microfiber with snaps, drawstring, collar, and Ethernet port access; (d) the back of the sweater shown form-fitted onto the robot's body.

in Figure 8(c), were performed in order to determine the optimal thickness of the silicone skin. Four millimeters provided a balance between the force required to move the robot's body (and thus placing strain on the motors) and how much material was required for a soft feeling and minimizing visible gaps between the plates.

6.2.2 Clothing Materials. We prioritized the robot body's tactile quality as a way to invite touching the robot and promoting a sense of calm. We were initially drawn to sweatshirt material, given its softness and potential to symbolize clothing that is often worn at home while relaxing. However, sourcing sweatshirt material that could (a) repeatably stretch with the displacement of the robot's body in both the x and y directions, and (b) do so without loosening over time, was a challenge. Instead, we opted for a highly stretchable microfiber fabric. This fabric provided the mechanical qualities needed for the breathing motion as well as a soft exterior for inviting tactile interaction.

Another material consideration was the addition of grips on the robot's sweater. In internal experimentation with the sweatshirt material, we found that a user's hands could slip off the robot in certain positions when relaxing their arms. We explored a number of grip styles and materials (silicones, rubbers, foams, etc.); however, we found them to be sticky and cold as well as susceptible to peeling over time. Figure 9(b) shows one of these grip prototypes. Fortunately, switching to the microfiber material and utilizing fabric seams on the shoulders provided enough friction for users to place their hands without slipping.

6.2.3 Clothing Shape. We decided that a tightly fitting sphere shape for the covering would be able to best showcase the robot's breathing. In order to provide this spherical covering, we explored various methods of seaming the fabric together (Figure 9(a)). We opted for a series of seams that would mimic those of a human sweater—recessed horizontal seams at the robot's "shoulders" and side seams that outlined where sleeves would theoretically be (seen on the sides of Figure 9(c)). This approach provided a form-fitting spherical shape while also helping the covering appear more as clothes than skin. The addition of a small collar helped further the distinction of the sweater as clothing (visible in Figures 1 and 9(c)).

6.2.4 Attachment Mechanisms. Given the high stretch forces on the covering, we considered multiple ways to ensure that the fabric and silicone would remain in place during the robot's breathing motion. The silicone is thus mechanically fastened to the robot body with a custom clamp and fasteners (visible in Figure 8(c)). The fabric of the sweater is secured through a series of snaps in the back, a cinch tie at the bottom of the sweater (visible in Figure 9(c) and (d)), and an additional screw-on base plate at the bottom of the robot. The combination of these mechanisms provided ease of removal for washing, the ability to tighten the sweater against the robot body, and extra assurance that the sweater would not become unsecured during breathing.

7 Design Process: Digital Interactions

As we finalized the robot's form factor and material quality, we began to explore the ways a user would interact with the robot as an agent. These elements included: capacitive touch, eye animations, head motions, and audio effects. These elements were of extra importance given that we intentionally chose to omit natural language from the robot prototype. This choice was made for multiple reasons. First, we suspected, based on related work and the clinical researcher feedback, that the haptic interaction with the robot would be able to provide enough *guidance* to the user. Imbuing the robot with speech could also possibly detract from this haptic experience. Second, natural language processing can be difficult in home environments with ambient or extraneous noise, and research with voice assistants shows a propensity for frustration when devices misunderstand user input [62]. For an individual in a current state of high anxiety, a robot that cannot understand them easily has the potential to aggravate this anxiety state. Thus, natural language input to the robot was omitted from the design.

Luckily, the impact of non-verbal cues in social robots (through elements such as facial expressions, body movements, eye gaze, haptics, and audio) has been well studied and shown to contribute to trust, understanding, engagement, and task performance with robots [18, 78]. Specific to our design goals, we sought to imbue the robot with enough personality through non-verbal cues to reach our *engagement* design goal (which we suspected would in turn support our *instruction* goal), while also having them feel natural and non-threatening in order to achieve our *calming* goal. We additionally sought to optimize these digital interactions in a manner that encourages the caretaker mentality and empathy observed in previous testing.

7.1 Head Motions and Eye Animations

To reduce mechanical complexity while introducing more opportunities for robot expressions, we chose to implement a single degree-of-freedom in the robot's head. This enabled up-and-down nodding motions in the robot, which could be paired with audio effects and eye animations. We additionally did not want the robot to appear overly *anthropomorphized*, as discussed in Section 5, so a second degree-of-freedom was avoided.

The presence and activity of robot eyes are known to be some of the most important elements of social robot design [1, 13, 67, 86, 99]. Few studies have been performed specifically on eye design



Fig. 10. Examples of eye expressions for different robot states. From left to right: *Sleep*, *Idle*, *Celebratory*, and *Wake Up* (left two images).

for social robots, though both Onuki et al. [66] and Komatsu and Kamide [46] have found that larger, rounder irises are preferred. Onuki et al. report that participants found this type of eye shape as friendlier and better for sustaining gaze.

We experimented with multiple pupil and iris shapes, sizes, and colors within this large, round iris approach. While Zhu et al. [104] recently found that more complex eye designs increased trustworthiness, we found through visual inspection that a centered colored iris overly humanized the robot in comparison to the rest of the design. We therefore decided to move forward with a simple single black iris. However, even this "doe-eyed" circle was disconcerting when placed in the middle of a circular eye screen. This design gave off the appearance of the robot staring very intensely at the user. Instead, our solution was inspired by the googly eyes utilized in our earlier foam mock-ups (Figure 7), which were much-loved by early testers. When dropping the doe-eye to the bottom of the screen, we discovered that this style, particularly when combined with the upward angling of the head, gave the robot the sense that it was looking at users without staring at them uncomfortably. This style had another advantage: by not featuring particularly human-like eyes, we could "break the rules" and render a number of easily implementable eye expressions more similar to cartoon eyes (Figure 10). This enabled the eyes to take on more types of facial expressions, which was especially important given the robot's lack of an obvious nose and mouth.

7.2 Audio Effects

Given the intentional omission of natural language capabilities, we approached the design of *Ommie*'s audio effects as a core component to expressing the robot's emotional states. Past studies have demonstrated that audio alone can express robot emotions, and combining it with motion can enhance this effect [39, 49]. We accordingly designed two sets of audio elements: (a) chimes that would signal different phases of the breathwork (inhale, hold, and exhale) and (b) sound effects paired with eye animations and head motions to exhibit different robot emotional states.

The chimes were inspired by existing mobile applications for meditation and past experiences with live meditation sessions featuring traditional instruments such as singing bowls. We created the chimes sequences in GarageBand software and timed them to correspond to the lengths of each breath phase for various deep breathing cadences. We imagined this approach would provide the most clarity on when to transition between inhaling, holding, and exhaling in combination with the robot's haptic guidance. When testing two pitch variations, users had a preference for higher-pitched chimes.

The emotional state audio required a more nuanced approach. We sought to provide the robot with a sense of agency through sound that matched its visual level of anthropomorphization. This meant designing human-analogous sounds, rather than mechanical or animal-like, without being directly human. We took inspiration from robots in popular culture (e.g., Wall-e, R2D2) and commercial robots (e.g., Anki's Vector robot) that successfully find this balance. We experimented with sounds for *Ommie* by modifying open source recordings of (primarily adult) human sighs, coos, giggles, and yawns with the open source audio-processing software Audacity. Generally, speeding up the recordings and adjusting them to be a higher pitch provided audio that balanced between

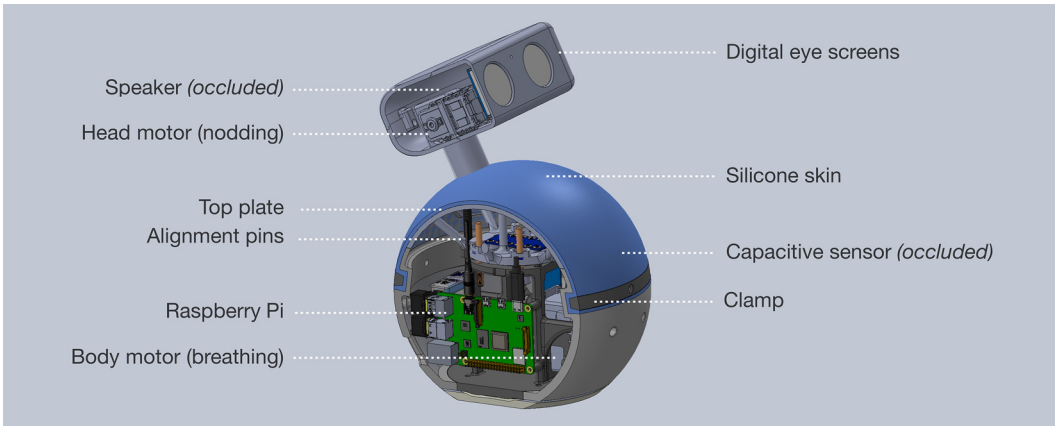


Fig. 11. CAD model of the prototype with labeled components. The moving top shell is constrained by a series of tracking pins as well as the silicone skin, which is bound to the adjacent front and back plates with a clamping mechanism.

being recognizably human-like (in a baby- or child-like way) while remaining not fully human. We hoped that these types of sound effects would make the robot approachable and encourage a care-taking mentality. Responses in both informal user testing with robotics graduate students new to the robot, and formal testing with participants (Section 9), were wholly positive in this regard.

7.3 Capacitive Touch

Without voice commands or a companion mobile or tablet application, we needed a way for users to communicate intent to the robot. This intent could be to wake up the robot, change the robot's setting, or start a breathing session, depending on the robot's state. We chose a capacitive touch sensor for this type of communication, given its inherent ability to directly invite touching the robot. This tactile invitation was an important element of the robot's interaction design that we had learned about in testing with the foam mock-ups. In order to test the capacitive interaction modality, we built a "wake up" sequence featuring robot head movements, eye animations, and sound effects. This sequence was triggered by an individual rubbing the robot's belly, where a capacitive sensor was hidden underneath the robot's covering (see Figure 11 for placement location). In testing this interaction informally with six individuals, they found the interaction natural and comfortable enough to include in the full robot prototype.

8 Full Prototype

With foundational design decisions made, we proceeded to build a fully integrated hardware and software prototype. This prototype was intended for use in a more formal and extensive participant design study with the public (Section 9).

8.1 Mechanical and Electrical System

The prototype's body is formed from a series of Multi Jet Fusion 3D-printed shells [16]. This manufacturing method was chosen for its robust material properties for mechanisms subject to high force loads. The top shell of the robot is attached to a Dynamixel MX-64AT motor via a linkage mechanism to produce the robot's up-and-down breathing motion (Figure 11). Each breath consists of approximately 2/3 of an inch of vertical displacement. The moving top shell is constrained by a series of tracking pins as well as silicone skin bound to the adjacent front and back plates with a

clamping mechanism. The silicon skin is made with a custom 3D-printed mold vacuum-cast with Smooth-On Dragon Skin silicone rubber. A flexible textile capacitive touch sensor is located on the front plate of the robot, underneath the silicone skin and robot sweater.

The robot head and neck were formed by Selective Laser Sintering 3D printing and feature one degree-of-freedom (nodding up and down) using a Dynamixel AX-12A motor. The head also houses a USB speaker and two 2.2" TFT digital screens for the eyes. The faceplate of the robot features a small circular cutout that was originally meant to expose a camera for more complex digital interactions (best seen in Figure 10). While we chose to omit the camera for this prototype, we kept the cutout as multiple users during informal testing commented on it as a way for the robot to theoretically inhale and exhale (as one user stated, "how else will Ommie be able to breathe?").

8.2 Software Controls

The robot is primarily controlled via a Raspberry Pi 4B, either directly in Python or with robot operating system. This system controls all I/O components, including the screens (via I2C protocol), capacitive touch sensor (via GPIO), speakers (via USB), and motors (via a Dynamixel U2D2 board converting USB to TTL signals).

We designed several robot behavioral states and transitions that combine these I/O components using a finite state machine. They include:[2]

- *Sleep*: Randomizing between three types of "sleep breaths" with audio coos and drooped eyes. This sleep state was designed to create the feeling that the robot was powered on and alive, even if it was not actively engaging in deep breathing.
- *Wake Up*: A transition behavior with audio and eye animations when the capacitive touch sensor is triggered. When a user rubs or pats the robot's belly, the robot proceeds to coo then slowly squints its eyes open (as shown in the right two images in Figure 10). This interaction ideally introduces haptic touch with the robot in a low-stakes manner and provides the robot with a display of agency.
- *Idle*: Micro eye movements while waiting to start a breathing exercise. This state aims to prevent users from feeling like the robot is frozen or overtly staring while waiting to begin the deep breathing interaction.
- *Breathing*: Moving the body upward and downward with closed eyes and audio chimes. This state is triggered via wireless secure shell protocol with specific breathing parameters (e.g., breath length, number of breaths). We designed the system to be customizable to allow the robot to perform any variation of deep breathing. For our usability study, we chose to use a 3-2-3 breathing (3-second inhale, 2-second hold, 3-second exhale) based on the procedure used by the clinical researchers we consulted with early in the design process.
- *Celebration*: Looking up at the user with joyful eyes and sounds before returning to *Sleep*. Providing a reward mechanism is an important part of developing a new behavioral habit [97]. We sought to use this state as a way to provide this reward via (a) helping the user acknowledge their success in completing deep breathing, and (b) portraying that the robot is happy and supportive of the user doing so.

9 Usability Study

We evaluated our full prototype with a formal usability study [55]. Our goals included: (a) determining if we met our design goals, (b) measuring the robot's ability to reduce anxiety states, and (c) gathering qualitative feedback on the robot interaction and imagined future use. All participant interactions were approved by our local Institutional Review Board. The following Sections 9.1

[2]Recordings of these states are available in the supplementary video.



Fig. 12. The study setup with a participant deep breathing with *Ommie* in their lap at the local university wellness center.

through 9.4, as well as 10.2, were originally reported in Matheus et al. [55]. In this article, Section 9.2 includes additional details on study methodology choices.

9.1 Participants

We performed the study with two participant cohorts: a *wellness* cohort and an *anxiety* cohort. The *wellness* cohort consisted of 21 participants (12 females and 9 males) recruited through a local university wellness center. Participant ages ranged from 18 to 38 years, with an average of 21.2 years. Fourteen participants (67%) had experience with deep breathing, with six (29%) having an active deep breathing practice at least once a week (of which three practiced every day). The *wellness* cohort was not screened for anxiety, and their results provided insight into any necessary study changes prior to testing with a more vulnerable population (those struggling with anxiety).

The *anxiety* cohort consisted of 22 participants recruited through the same wellness center and reported working with a therapist or doctor on anxiety reduction. Twelve identified as female, six as male, and four as non-binary. This male-to-female ratio is consistent with reporting on the prevalence of generalized anxiety in the United States [92]. Participant ages ranged from 18 to 29 years (average 20.9 years). All but one participant had prior experience with deep breathing, though only eight actively practiced it regularly at least once a week. None reported practicing daily. This distribution was not unexpected given the known challenges of compliance with therapy-given practices [32].

9.2 Method and Measures

Each session lasted approximately 15–20 minutes in a private living room-like setting in the university wellness center (Figure 12). The interaction with the robot consisted of the participant:

(1) Being introduced to "Ommie" in its *Sleep* state on the table.
(2) Waking up the robot by rubbing its "belly" and observing its *Wake Up* sequence.
(3) Observing the robot's *Idle* state.
(4) Placing one's hands on the robot and being asked to match one's breathing to the robot's breathing. No information was provided to participants on the length of the breath phases or number of breaths. This choice was to ensure that the participants were only guided by the robot and not by pre-existing information.

(5) Breathing with the robot in a sequence of three 3-2-3 breaths (3-second inhale, 2-second hold, 3-second exhale) for three full breaths. This cadence was chosen as the easiest and most accessible version of the deep breathing cadence used by one of the clinical researchers while working with patients in their clinic. Past work with deep breathing HCI devices has also shown users can sometimes struggle to breathe slowly [53], which we sought to avoid.
(6) Observing the robot's *Celebration* and its return to *Sleep*.
(7) Optionally repeating steps 4–6 with the robot in one's lap.

For the *anxiety* cohort specifically, we measured pre- and post-interaction anxiety levels with a widely used short-form version of the State-Trait Anxiety Inventory, the STAI-6 [54]. This short format features six four-point scales (from "not at all" to "very much") on particular mood states (e.g., relaxed, worried, tense). It has been shown to be comparable to the full-form survey for measuring current anxiety levels [54]. Participants completed the STAI-6 immediately before and after interacting with the robot. We opted for this survey-based method for anxiety for multiple reasons. First, the STAI-6 is a gold standard metric in the psychological literature. Second, we considered more objective biometric sensors to measure anxiety (e.g., heart rate, heart rate variability, galvanized skin response) as less useful and potentially intrusive. Some of these sensors need to be placed on the hand or fingers, which would impede the haptic use of *Ommie*, and the addition of electronic sensors directly on the body may have increased anxiety in our users. Additionally, we considered the user's own internal feelings as the most important for the robot's success and the user's desire to use the robot again (thus contributing to our *engagement* goal).

An informational interview followed the robot interaction.[3] The interview began with open-ended questions on general impressions of the experience and continued with more targeted questions about topics such as the physical handling of the robot and the robot's role. Following the interview, participants submitted an exit survey with 7-point Likert scales of robot and interaction characteristics. These characteristics included: *calming*, *approachable*, *engaging*, *desire to interact again*, and *desire to use in the home* (with the latter characteristic only for the *anxiety* cohort).

9.3 Quantitative Results

Overall, results showed positive regard for the robot in both cohorts and a measurable drop in anxiety measures for the *anxiety* cohort. All participants were successful in using the robot from start to finish, and the exit survey results showed favorable rankings on all five characteristic scales (Table 1). Overlapping exit survey data were analyzed between the two cohorts using Mann-Whitney U testing.[4] Notably, no statistically significant difference existed between cohorts among the four shared characteristic scales. We additionally split exit survey data into three experience-level groups: those with no deep breathing experience (7), those with an occasional practice (22), and those regularly practicing at least once a week (14). Between these groups, all scales had statistically similar results apart from *engagement*. *Engagement* scores for all groups were high: no experience averaged 6.43 (0.49), occasional practice averaged 5.91 (1.08), and regular practice averaged 5.57 (0.90). However, those with no experience reported significantly more engagement than those with a regular practice ($p < 0.05$).

Anxiety cohort STAI-6 data were analyzed using Wilcoxon signed-rank testing (Table 2, Figure 13). All participants had improvement in at least one anxiety state post-interaction. Reduction in total STAI-6 anxiety measurement was significant ($p < 0.001$), averaging 3.59 points (−25.9%).

[3]The *wellness* cohort was offered the lap interaction during the interview.
[4]We used non-parametric tests as the exit survey and STAI-6 data were determined to be non-normally distributed with Shapiro-Wilk testing.

Table 1. Exit Survey Results (7-Point Scale)

		Calming	Approachable	Engaging	DTI[1]	DTH[2]
Wellness	Mean	6.05	6.19	6.10	6.24	n/a[3]
Cohort	SD	0.67	0.87	0.89	0.89	n/a
Anxiety	Mean	6.32	6.09	5.68	6.00	5.36
Cohort	SD	0.72	1.23	1.09	1.20	1.65
Δ	p	>0.05	>0.05	>0.05	>0.05	n/a
No	Mean	6.14	6.00	6.43	6.29	n/a
Practice	SD	0.64	0.76	0.49	0.70	n/a
Occ.	Mean	6.14	6.05	5.91	6.09	n/a
Practice	SD	0.81	1.26	1.08	1.12	n/a
Reg.	Mean	6.29	6.36	5.57	6.07	n/a
Practice	SD	0.45	0.72	0.90	1.03	n/a

[1] *Desire to interact again.*
[2] *Desire to use in home.*
[3] *"n/a" is listed since the "desire to use in home" measure was not collected for the wellness cohort.*

Table 2. STAI-6 Pre- and Post- Intervention (Anxiety Cohort)

		TA[1]	Calm	Tense	Upset	Relaxed	Content	Worried
Pre	Mean	13.86	2.45	2.23	1.36	2.14	2.64	2.50
	SD	2.92	0.60	0.75	0.66	0.64	0.73	1.06
Post	Mean	10.27	3.23	1.59	1.05	3.00	2.86	1.73
	SD	2.37	0.61	0.59	0.21	0.76	0.77	0.88
Δ Pre and Post	Mean	−3.59	0.77	−0.64	−0.32	0.86	0.23	−0.77
	SD	2.40	0.43	0.66	0.65	0.83	0.61	0.75
	p	<0.001*	<0.001*	<0.001*	<0.05*	<0.001*	>0.05	<0.001*

[1] *Total anxiety, calculated with standard positive state reverse-scoring.*
[*] *Statistically significant.*

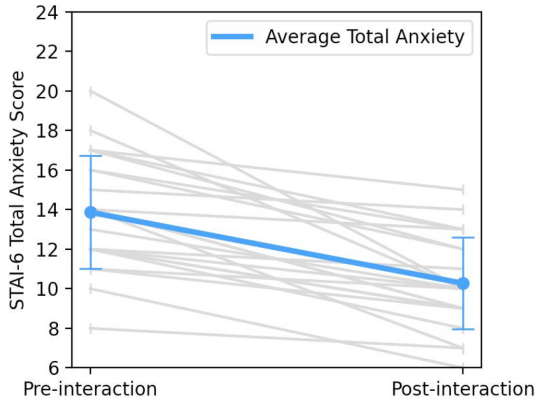


Fig. 13. STAI-6 total anxiety measures for the *anxiety* cohort.

9.4 Qualitative Results

We analyzed information from the participant interviews by transcribing written notes and applying a digitized affinity diagram technique [50] to distill patterns across individuals. Analysis revealed five core insight clusters.

9.4.1 Participants Felt Comfortable with the Robot and Successfully Deep Breathed with It. All 43 participants (including those without deep breathing experience) were observed to successfully breathe in the same cadence as the robot's haptic instruction. All participants were comfortable with the concept of placing their hands on the robot. A few individuals described the first moments of feeling the robot's breathing as "strange" or "odd" but that they enjoyed it by the end. Nineteen *anxiety* participants (86%) closed or lowered their eyes when breathing with the robot, potentially a gesture of trust.[5] All 43 participants also opted to try the more intimate option of having the robot in their lap. When asked about their overall experience with the robot, many users commented on how "natural," "easy," and "intuitive" it felt. The reasons behind these sentiments centered around (a) the rise and fall of the robot's breathing making sense, (b) the robot's endearing form factor and soft materials, (c) the robot's personality (primarily the eye animations and sounds), and (d) similarities to other calming experiences, such hugging a friend or pet.

9.4.2 Participants Successfully Tactilely Interacted with the Robot in a Wide Variety of Ways. Across both cohorts, we observed seven different variations of user hand placements with the robot on the table and 10 variations when using it in a lap (Figure 14). Multiple users shifted their hands during the interaction to find their ideal position. The preferred interaction method varied: 23 participants preferred the lap, 12 preferred the table, 4 preferred the lap with certain modifications, and 4 had no strong preference. Those who preferred the lap cited physical comfort, better feeling of the robot's breath, and the desire for a more "intimate" interaction. As one participant explained, "[Ommie] felt alive, and I wanted to be closer." Those that preferred the table interaction similarly cited physical comfort and feeling more of the robot's breath. They also noted wariness around the fragility of the robot in the lap, or that the table use felt more "filled with intent." All in all, participants had clear individual preferences about how to use the robot, and the robot was able to meet them.

9.4.3 The Haptic Interaction Provided Focus, Guidance, and Connection. The vast majority of participants commented on the unexpected benefits they received from the haptic breathing interaction with the robot. Participants described it with such terms as "focusing," "grounding," "anchoring," and "tethering." A common reason for this focus was the ability to tune out of thoughts and other distractions by way of the physical sensation. As one participant noted, "I could let my hands guide me, not my thoughts." The easy-to-follow nature of the haptic guidance also contributed to feelings of focus. One participant elaborated, "You don't have to process [the deep breathing]…it's just there. You know when to do it."

Multiple participants also preferred the non-verbal guidance (primarily the haptics and secondarily the audio chimes) to prior meditation experiences. One participant explained: "one problem with meditation is focusing on words and the voice, over focusing on yourself. With Ommie…I was focusing less on Ommie and focusing more on myself." Other participants commented on the haptic feature providing a closer relationship with the robot. Many compared the experience to "embracing" or close contact with a friend, and one participant shared that "having a sense of touch and feeling the motions of the robot brought a deeper level of connection."

[5]The eyes closed measure was not formally collected in the *wellness* cohort.



Fig. 14. Examples of various participant hand placements during haptic interaction with the robot in the usability study.

9.4.4 Participants Saw the Robot's Role as One of Helpful Companionship. While the exact terms they used to describe the robot's role varied, all but one participant in the *anxiety cohort* and the majority in the *wellness* cohort spoke of the notion of companionship. This manifested in descriptions of the robot as "pet"-, "friend"-, "child"-, or "companion"-like and the feeling of participating in an activity together. For instance, one user noted, "there was a tangible feeling of someone doing this with you," and another described feeling "less lonely" than their solo practice. Multiple participants drew an analogy to the motivating social aspect of group meditation but without the experienced drawbacks of feeling judged or being distracted.

Notably, 11 *anxiety* cohort participants (50%) spoke of a dual role of companionship combined with a more intentional and skilled level of help or instruction (such as from a "teacher," "professional," "coach," "guide," or "therapist"). Six participants specifically noted the duality with a pet-like companion, with statements such as: "like sitting in a therapist's room, but the therapist is a little cute animal." These responses combined two relationship hierarchies between humans and robots: one of receiving top-down instruction from the robot, and another of taking care of the robot. When asked, one user found the hybrid relationship strange, but the majority stated they were unphased by the combination of two distinct roles. Instead, participants expressed the advantage of being guided without feeling commanded or bossed around. Rather, as one user put it, that the robot is "someone trying to help me."

9.4.5 Participants Could Imagine Using the Robot at Home and How It Might Solve Existing Challenges with Deep Breathing Technologies. From the *anxiety* cohort exit survey, 16 participants (73%) indicated a clear interest to use the robot in their home, 4 (18%) had little to no interest, and 2 (9%) were somewhat interested. Those with high interest described the robot as solving certain challenges they had faced with starting or sustaining a deep breathing practice: (a) remembering to practice (particularly when in a high anxiety state), (b) keeping count of their breathing, (c) time spent searching for audio guides or other media, (d) getting distracted on their phones when using mobile apps, and (e) feeling "alienated" or overly "commanded" by a voice. They saw the robot meeting these challenges based on its physical presence, haptic interaction, ease of setup and guidance, and companionship. Five *anxiety* cohort participants (23%) commented on another benefit vs. existing techniques: faster relaxation time. As one participant exclaimed, "I relaxed way quicker than my normal 5-10 minute exercises!"

The four (18%) participants with little interest in home use cited various reasons. Two participants had pre-existing proficiency in deep breathing and did not feel a need to use the robot as a result. One of these participants also commented they may be embarrassed if someone were to walk in on them using the robot. One participant commented that they would use the robot at home if the robot head were merged into the body, making it easier to cradle. Two participants also were uncertain of the value of the robot in comparison to the ease and portability of a mobile application.

10 Discussion

10.1 Design Goals

A core part of our usability study was to validate meeting our design goals. The following provides an analysis of why we believe we met our four goals, based on our quantitative and qualitative measures. We additionally reflect on what design choices may have led to these results.

(1) *Calming*: STAI-6 measures showed a statistically significant reduction in anxiety measures before and after using the *Ommie* robot. Of all measured anxiety states, *calm* and *worried* had the most positive change, showing strong calming effects. Participants additionally rated the robot to be highly calming (an average rating of 6.05/7 for the *wellness* cohort and 6.32/7 for the *anxiety cohort*). Participants noted the positive calming effect that the robot had on their interviews as well, noting a clear difference after interacting with the robot. We suspect that the robot morphology and haptic interaction were the primary design drivers. The morphology of the robot itself was identified by many users as cute and approachable, likely due to its minimal and bulbous form factor, smaller scale, doe-eyes, and child-like audio chirps. The choice to utilize a soft surface also likely provided a pleasant and calming feeling for users. We noticed, for instance, some participants spontaneously rubbing parts of *Ommie*'s belly to feel the material.

(2) *Engagement*: Participants rated the robot as highly engaging on survey Likert scales (average 6.10/7 for the *wellness* cohort and 5.68/7 for the *anxiety* cohort). They additionally commented on their enjoyment of interacting with the robot in the interviews, and many expressed a desire to use the robot again. We see the fact that all participants opted to use the robot a second time as also supporting the robot's capacity for engagement. We note, however, that these measures only capture engagement from a single session with the robot. Future study is required to measure long-term engagement with repeat use. For this single interaction, we suspect that certain design choices led to high engagement results. For one, most users did not expect to haptically interact with the robot, and this modality may have provided more engagement opportunity than normal. Many users also commented on *Ommie*'s audio chirps and eye animations as effective at emotional expressivity despite their minimal designs, likely leading to a more engaging interaction.

(3) *Inviting Haptics*: All participants were able to feel the haptic interaction clearly and all were comfortable with touching the robot. Participants were also able to customize their haptic interaction with the robot by holding it in a wide variety of ways. In the interviews, many commented on the appealing softness of the robot and the positive impact that the haptic interaction had on their ability to focus. We suspect that our purposefully designed "wake up" sequence for the robot, as triggered by a user rubbing *Ommie*'s belly, helped users get comfortable with tactilely interacting with the robot. It is also likely that our morphology and scale testing paid off, with the fully spherical design (akin to a sports ball) and slightly smaller, more approachable scale making it easy for users to know where to place their hands.

(4) *Breathing Instruction*: All participants were able to match their breathing to the robot's breathing successfully, and they were able to do so with minimal facilitator instruction. No participant expressed trouble with understanding how the robot was guiding them in the deep breathing exercise. Based on the qualitative interviews, there were two primary design decisions key to achieving this goal: the haptic expansion and the audio chimes. We suspect that the combination of the two is more impactful than just one or the other.

10.2 Study Insights

In addition to validating our design goals, we discovered certain insights about the robot's interactions through the usability study. We had expected certain results based on past research in the field, such as the impacts of robot personalities [24] or the calming effects of touch [80]. However, there were a few surprising findings:

10.2.1 Haptic Focusing Effects. First, we did not expect the haptic interaction to provide such a focusing experience in addition to a guiding one. One could have expected those with regular breathing practice to be impervious to the robot's guidance or, for those without one, to lack the experience required to become so quickly focused. Instead, participants from varying experience levels among both cohorts reported the focusing effect. We suspect that this effect results from simultaneous sensorial enhancement and deprivation. Tactilely feeling the breathing guidance allows users to block out the visual world and reduces the need to be auditorily present. It additionally allows participants to tune out their conscious mind by providing a new stimulus on which to focus. Given that current technologies designed to support deep breathing (e.g., mobile applications, recordings, and certain robots) do not provide tactile feedback, we see *Ommie's* focusing effects as a unique characteristic.

10.2.2 Tactile Versatility. Second, we were surprised at the multitude of ways participants held the robot on a table or lap. In our mock-up testing, users always placed their hands on the robot's "shoulders." We suspected this felt like the most appropriate way to handle an anthropomorphized robot with a certain level of agency, and we expected the same from study participants. Results showed otherwise, with one participant statement suggesting a possible reason why: "[the lap use] felt very intimate at first…am I invading something's personal space? But, Ommie seemed to want it." We theorize that study participants felt comfortable customizing their hand placement because of designed invitational features: (a) an introductory haptic experience (rubbing the robot's belly), (b) the robot's sweater making it both soft and seemingly clothed (vs. a distinctly un-clothed foam mock-up), and (c) the eye and audio expressions of the robot leading to the oft-stated *pet*- or *child*-like nature and affording caregiving touch.

10.2.3 Companion Role. Finally, we were not certain how participants would interpret the robot's role during the interaction. Notably, nearly all participants described it as one of companionship. One could imagine those with deep breathing experience seeing any sort of guidance as over-instruction, or those struggling to practice seeing mere companionship as not enough. We find the companionship result potentially beneficial, however, given that teacher- or coach-like relationships can sometimes introduce stress or pressure. These roles inherently involve some level of judgment or testing. Interestingly, with *Ommie*, the 50% of participants using terms such as a "coach" or "teacher" did so in tandem while also describing a companionship component. We see this as evidence of a balancing act for mental health and HRI: providing enough guidance so that users know how to do a behavioral practice correctly, but with a gentle enough touch to retain feelings of closeness and partnership while avoiding those of judgment. We suspect that users felt this balance partially due to the robot's friendly, approachable nature (e.g., its audio and eye expressions, soft body, and smaller size). It is also likely that the haptic element made the interaction feel more intimate and thus more similar to encounters with close friends or pets.

10.3 Limitations

While the *Ommie* robot shows promise, our work is not without limitations. With regard to our design process, much of the early work for the robot was performed during the COVID-19 pandemic, so access to early user testing was limited, and anxiety levels were abnormally high. As such,

we used clinical researchers as proxies for need-finding, and we used primarily non-vulnerable individuals for iterative testing. By waiting to involve users with diagnosed anxiety until later on, we could have missed pieces of context information or user needs early on. Our expert interactions were also limited to two individuals, and other experts may have had more diverse feedback. Additionally, given the wide array of design decisions to be made, some were made with more minimal testing than others (e.g., a small number of users, users more experienced with robots, only internally testing ourselves). While our full design was ultimately validated and performed positively with users with diagnosed anxiety, any of the prior limitations could have skewed our early design decision-making or missed a concept that would have had more impact for them.

With regard to our usability study, there are a few limitations. First, the wellness setting of our study featured a quiet and calm atmosphere, potentially contributing to reported feelings of calm and focus with the robot. Similarly, the way the robot interaction was facilitated by an experimenter, in a calm and approachable manner, could have also had an effect on our results. Third, robot studies are notoriously affected by *novelty effects* [48], which likely played a role in our findings, particularly with measures of engagement. Our measures may have also been influenced by the fact that testing sessions were relatively short per participant. We suspect longer-term, in-the-wild use will bring additional challenges (e.g., sustained engagement). Lastly, another limitation of the user study was the focus on young adults. Results may vary in other populations.

10.4 Future Work

The design process and usability study described in this article represent the first steps in creating a platform for supporting deep breathing and mental health with a social robot. Future work surrounds studying the robot in longer-term, autonomous deployments and exploring methods for maintaining compliance to deep breathing over time (particularly given prior research showing difficulty in long-term engagement with robots and the novelty effect) [23, 48]. One particular area of interest is how *Ommie* can work in tandem with existing therapy practices. Informal impressions with the original clinical researchers from Section 4.1 on the final *Ommie* prototype were positive, but exploring the receptivity and relationship of the robot in clinical settings is warranted. Deploying a robot like *Ommie* through established structures and systems for mental health care offers an existing pathway to our target user group, and could help bolster engagement and trust in the robot if introduced by a clinician. At the same time, users may feel like they are being watched by the clinic via the robot, particularly if they use the robot at home. Or, if used in the clinic by multiple users, they may feel less emotional connection with the robot or at risk of judgment from others. These remain important open questions for future study.

It should be noted that deep breathing has a number of therapeutic benefits beyond anxiety. It has been shown to help individuals with depression [81], those with increased stress [72], as well as those with asthma [77]. Deep breathing is also often a component of other mindfulness practices, such as meditation and guided imagery. These domains provide many future therapeutic applications of the *Ommie* robot. Understanding what applications are appropriate for *Ommie*, and how the robot performs in comparison to non-robotic interventions (e.g., deep breathing guided by an audio application, typical therapy sessions), are other areas for deeper investigation.

11 Conclusion

In this article, we presented *Ommie*, a novel socially assistive robot that reduces anxiety by supporting regular deep breathing practices. We built upon prior research in both HRI and HCI for supporting mental health and contribute a new robot to this growing field. We additionally drew from insights in how social robots have previously been able to support therapeutic calming effects through haptic interactions as well as teach users new skills. We implemented a user-centered

design process with extensive iterative prototyping across sketching, clay modeling, foam mock-ups, digital creation, and mechanism design. Our design process was informed by collaboration with clinical researchers in anxiety, who shared their experiences and research in psychology and neuroscience about the calming effects of haptics and the known anxiety-reducing effect of practicing deep breathing. Thus, our robot featured a haptic interaction (i.e., feeling the robot tangibly breathe) that is not currently present in social robots that aim to support behavioral practices for mental health. The robot additionally featured a semi-anthropomorphized form factor that draws from typical morphologies seen with robots that teach new skills, while imbuing the minimalism and tactile qualities seen with robots that support therapeutic calming. Further core design decisions discussed include: robot breathing motions, textiles, eye animations, audio effects, and touch interactions.

We investigated whether our robot prototype met our original design goals (*calming*, *engaging*, *haptic interaction*, and *instruction*) and reduced anxiety states in a two-cohort usability study [55]. Results using a standardized anxiety states measure (STAI-6) showed a significant overall reduction in anxiety states after use with *Ommie*. Users perceived the robot as calming, approachable, and engaging, with the majority indicating a desire to use the robot again. Interestingly, individuals with no experience deep breathing reported significantly more *engagement* than those with a regular practice. These results, along with informational interviews, provided validation for meeting our design goals as well as a number of surprising qualitative findings. For instance, participants varied widely in how they tactilely interacted with the robot. They additionally commented that the haptic interaction provided a better ability to focus (and often faster time to focusing) in comparison to previous experiences with verbal deep breathing guidance. Participants also described their relationship to the robot as one of companionship, with half of participants noting a hybrid role that combined companionship with a more formal coaching or teaching relationship. Our work with the *Ommie* robot shows the capacity of a social robot to support deep breathing, a cornerstone behavior for anxiety reduction and mental health. We hope that this work inspires and invites further study into the nascent field of social robotics for mental health.

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